



Suspicion of online product reviews as fake: Cues and consequences

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ABSTRACT

Consumers rely on online reviews because they deem information from a third party to be more credible than promotional communications. Unfortunately, not all online reviews are legitimate. It is therefore critical for marketers to gain a better understanding of how fake online reviews affect consumer purchasing behavior and what cues consumers use in evaluating review veracity. The current research examines the effect of suspicious reviews on reviewer's opinion valuation, consumer attitudes toward the brand and website, and purchasing. Then it identifies the specific cues that consumers use to evaluate whether a review is fake or credible.

1. Introduction

Online sales represent a rapidly growing proportion of total retail sales. Word-of-mouth (WOM) increases consumer awareness and leads to product sales (Li & Hitt, 2008). Consumers rely on WOM as a source of information, especially when it comes to evaluating experience goods (i. e., products not easily evaluated until after the product is used) (Li & Hitt, 2008). With the explosion of the Internet, WOM has moved from interpersonal to national and global networks (Avery, Resnick & Zeckhauser, 1999).

Online product reviews have been around for about two decades (Pranata & Susilo, 2016). It was in 1999 that Avery, Resnick and Zeckhauser determined it was possible to share product evaluations via the internet. Since that time, the creation and use of online reviews have grown exponentially (Pranata & Susilo, 2016).

Online reviews have a determining effect on product evaluation and purchasing. According to research conducted by Dixa in 2022, 93% of online shoppers read the reviews and 97% of their buying decisions are affected (Loiselle, 2022). Consumers are more likely to rely on online reviews because they feel the information presented by manufacturers is biased and less credible (Bansal & Voyer, 2000; Smith, Menon and Sivakumar, 2005). Consumers rely on trust and credibility as key aspects in online product evaluation and decision making. Trust and credibility relate to perceptions of reliability and honesty (Gössling, Hall & Andersson, 2016).

However, not all online reviews are perceived as credible. Fake product reviews are defined as “deceptive reviews provided with an intention to mislead consumers in their purchase decision making, often

by reviewers with little or no actual experience with the products or services being reviewed” (Zhang et al., 2016). E-commerce is being increasingly challenged by fake online reviews (Sussin & Thompson, 2012).

In this article, we raise two critical research questions. First, what are the attitudinal and behavioral consequences regarding decision making and purchasing when consumers are suspicious of online reviews? This first question establishes the importance of investigating the effects of fake online reviews. Second, what are the cues that consumers use to evaluate whether an online review is fake or credible? This second question seeks to explain what leads consumers to make such an evaluation, thereby revealing specific and appropriate managerial implications.

This study hopes to make several significant contributions to the existing literature. First, we explicitly consider the initial impact of reviews perceived as fake on the valuation of the reviewer's opinion and subsequently on the remaining outcomes of attitudes toward the brand, attitudes toward the website, and purchase completion. This allows for a more complete understanding of the mechanism by which fake reviews ultimately affect purchasing decisions, exploring the effect on the manufacturer (via brand perceptions) as well as the reseller (via website perceptions). Furthermore, we explore the notion that the cues used by consumers to evaluate a review as fake are not the same cues as those used by consumers to conclude that a review is credible. Concurrently, we seek to rule out those cues that are not actually used by consumers in their assessment. The findings provide useful insights for brand and website managers, as well as for market researchers.

Our research is organized as follows. First, we provide the theoretical

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background for the current investigation. Second, we present a review of the literature regarding online product reviews, fake online reviews, the consequences of fake online product reviews, and the cues used by consumers to evaluate a review as fake versus credible. Next, we conduct two studies to investigate the research questions and report our findings. To conclude, we discuss theoretical contributions, managerial implications, limitations, and future research.

2. Theoretical background

Relevant theoretical bases for the current research include persuasion knowledge, referral congruity and motive attribution (see Putz, 2017). First, according to the Persuasion Knowledge Model (Friestad & Wright, 1994), consumers develop a specific persuasion-related knowledge structure that helps them evaluate persuasion attempts (such as online reviews). This knowledge structure is activated (Fein, Hilton & Miller, 1990) when consumers become suspicious of reviewer motives, questioning the authenticity of the reviewer and the review (DeCarlo, 2005).

Second, suspicion of online reviews may be triggered when a consumer perceives a lack of congruity between the reviewer and the product. When consumers are presented with unexpected or implausible behavior, they engage in greater information processing (Lee & Schumann, 2004) and scrutinize the reviewer's underlying motives (Pyszczynski & Greenberg, 1981). Putz (2017) empirically demonstrates that a lack of perceived congruity leads consumers to be suspicious of the authenticity of an online review.

Finally, consumers differentiate between intrinsically motivated, credible reviews (those designed to help other consumers) and extrinsically motivated reviews (often *perceived* as being associated with some form of incentive). When motives are perceived as extrinsic, the consumer reacts to the reviewer and negatively evaluates the brand as a result (Campbell, 1995). Putz (2017) reports that perceived intentions affect attitudes toward the brand as well as purchase intentions. In summary, the consumer's persuasion-related knowledge structure is activated by a perceived lack of congruity between the reviewer and the product, leading consumers to question the honesty of the product review as well as the credibility of the reviewer. Thus, suspicion results from consumer perceptions.

3. Literature review

In the following sections, we discuss the existing literature specific to (1) online product reviews, (2) fake online product reviews, (3) consequences of fake online reviews, and (4) cues used by consumers to evaluate reviews.

3.1. Online product reviews

When consumers initiate the purchase decision-making process, they engage in several stages, one of which is identification and review of product information (Frank, Massey & Lodahl, 1969). Customers browse internal and external sources and choose information to rely upon (Bunn, 1993). Accordingly, the purchase decision can be greatly influenced by consumers' choice of information and the way they interpret signals from the chosen information (Lee, 2018).

During the online shopping process, consumers rely almost exclusively on online product reviews (aside from personal recommendations) to reduce uncertainty and make more informed choices (Chakravarty, Liu & Mazumdar, 2010; Hugstad, Taylor and Bruce, 1987). Online product reviews can be described as peer-posted product evaluations uploaded on company or third-party websites (Lee & Choeh, 2016).

Most product web sites use some form of star rating system in addition to product reviews. However, as demonstrated by Lee and Choeh (2016), written reviews are more helpful than ratings in

supporting purchase decisions. With a star rating, users cannot determine the "real semantics" of statements or the reviewer's feelings, experiences and opinions on the product (Lee & Choeh, 2016). Factors affecting assessments of product quality and performance include, for example, the reviewer's reputation, the disclosure of the reviewer's identity, the depth of the review (Lee & Choeh 2016), and the inclusion of both pros and cons (Filieri et al. 2018).

3.2. Fake online product reviews

According to the report *Fake Online Reviews 2021*, fake reviews are defined as "any positive, neutral, or negative review that is not an actual consumer's honest and impartial opinion and does not reflect a genuine experience of a product, service or business." Deceptive reviews intend to mislead consumers (Zhang, Ko & Carpenter 2016). Despite the commitment of review platforms to combat fraud, about 15–30% of reviews are fake (Belton, 2015; Luca & Zervas, 2016; Sussin & Thompson, 2012). More recently, according to Saoud Khalifah, Founder and CEO of Fakespot, while fake reviews generally account for 30%, prevalence increased to 35–40% during the Covid19 pandemic (Schoolov, 2020).

When consumers read online reviews, they often consider the motivation of the writer to post a particular review and decide whether the source has a stake in the action taken by the consumer after reading the review (Clare et al. 2018; O'Reilly et al., 2016). Zaman et al. (2023), who conducted an extensive investigation of individual motives for posting fake reviews, found that motivations may be intrinsic (e.g., hating the brand, hating the brand ambassador, retaliation) or extrinsic (e.g., monetary rewards, free or discounted products, coupons, enhancing self-esteem or opinion leadership). Consumers process the information from the review to arrive at a judgment regarding the credibility of the product review. For example, when a reviewer asks consumers to buy, click on, or forward the post, the reviewer's motivation may be deemed uncredible (O'Reilly et al., 2016).

Some product reviews are motivated by efforts to increase market share and are therefore instigated by companies, rather than ordinary consumers. Interestingly, Wu & Qiu (2023) provide empirical support reporting that sellers with high quality products are unwilling to support fake reviews, while sellers with low quality products are. One strategy that companies use to manipulate product reviews is called 'sock puppet' where companies create bogus accounts to write positive reviews of their own brands (Schoolov, 2020). Sellers can also hack into a customer's Amazon account and post a positive review without the customer's knowledge (Schoolov, 2020).

In other cases, businesses hire individuals to post either *fake positive* reviews for their products or *fake negative* reviews for competitive products, constituting review fraud (Dellarocas, 2006; Lappas, 2012; Luca & Zervas, 2016; Mayzlin, Dover & Chevalier, 2014). However, as explained by Zhuang, Cui and Peng (2018), adding fake reviews leaves more cues of manipulation, thus arousing consumer suspicion. If the objective is to alter the balance of positive versus negative reviews, an alternate strategy of deleting negative reviews is undetectable. That said, the absence (or lesser number) of negative reviews in and of itself could raise suspicion (Zhuang, Cui & Peng, 2018).

The practice of offering individuals some form of compensation (such as coupons, discounts, raffles, free samples, etc.) in exchange for writing reviews (Abendroth & Heyman, 2013) is extensive. He, Hollenback and Proserpio (2021) found that there are about 23 fake review-related groups on any given day, most of which are large and active, averaging 16 K members and 568 fake review solicitations per day. Cui et al. (2022) point out that although incentivized reviews may or may not be fake reviews, they still impact brand evaluations and purchase decisions. The determinant feature of perceived fake reviews is whether they mislead consumers (Wu et al., 2020), and it doesn't matter if such reviews are incentivized, positive, negative, or neutral (Sudhakaran, Hariharan & Lu, 2016).

Detecting fake reviews *online* is difficult because the cues available in face-to-face communication (tone of voice, facial expressions, gestures, etc., sometimes known as “tells”) that individuals use to assess whether someone is telling the truth do not exist (Donath, 1999; Wu et al., 2010). Therefore, online consumers apply “heuristic (or peripheral) processes and cues” to evaluate trustworthiness (Chaiken & Maheswaran, 1994). The challenge is that fake reviews are often intentionally designed to appear as authentic, making them difficult to detect (Lappas 2012). In any event, consumer knowledge, experience and expertise do play key roles in alerting consumers to manipulative intent (Kirmani & Zhu, 2007).

3.3. Consequences of fake online reviews

The immediate question is if and how the presence of fake reviews inhibits customer purchasing. One perspective presented by Zhuang, Cui and Peng (2018) is that fake reviews may not affect consumer evaluation and purchasing because consumers are aware that fake reviews exist and are sophisticated enough to discount and adjust their interpretations of online opinions. Alternatively, Pavlou & Gefen (2005) explain that fake reviews *can* negatively impact purchase because the consumer feels a sense of injustice. As a result, consumers conclude that products with suspected fake reviews are of lower quality, so they adjust their perceptions accordingly, leading to a decrease in sales (Zhuang, Cui and Peng, 2018).

If indeed sales are damaged by the presence of fake online reviews, what is the mechanism and what other important outcomes are affected? Based on existing literature, it appears that suspicion of a fake review directly leads the consumer to deem the reviewer uncredible. More specifically, once consumers suspect a certain review, they will begin to discount the value of the reviewer's opinion as insincere and untrustworthy (O'Reilly et al., 2016; Tuk et al., 2009). Not only are consumers likely to discount the recommendations of the reviewer, but the perception of the product quality is also affected to the extent that consumers disengage from the purchase process (Dellarocas, 2006; Mayzlin, 2006). Furthermore, perceptions of reviews as fake may transfer to the reseller website, for example, by concluding the reseller site lacks the necessary tools for effectively screening out fake reviews (Roman, Riquelme & Iacobucci, 2023). Thus, in the bigger picture, consumers may show distrust toward the review, the reviewer, the brand, and the website (Kim, Maslowska & Tamaddoni, 2019; Roman, Riquelme & Iacobucci, 2023; Sparks & Browning, 2011). However, the relationships between and among these constructs is not clear. Accordingly, the first research question to be addressed is:

RQ1: What is the mechanism by which fake online reviews impact consumer purchasing decisions?

3.4. Cues used by consumers to evaluate reviews

The content and features of online reviews offer cues for consumers to assess authenticity (Zhuang, Cui and Peng, 2018). Based on a thorough review of the literature, we propose that the cues used by consumers to evaluate an online product review as fake are different from those used by consumers to assess a review as credible. There are some cues (such as a reviewer asking a consumer to buy/click/forward) that if present suggest an online product review is fake (O'Reilly et al., 2016). However, the absence of being asked to buy/click/forward does not lead the consumer to conclude that the review is credible. On the other hand, the presence of a verified buyer badge, which affirms that the reviewer has actually purchased the product, leads the consumer to believe that the review is credible (Lappas, Sabnis & Valkanas, 2016), while the absence of this badge does not lead a customer to believe the review is fake. This leads to our second research question:

RQ2: What are the cues that consumers use to evaluate online product reviews as fake, (versus reviews that are credible)?

We next review the existing literature to distinguish those cues which

arouse suspicion that a review is fake from those which lead a consumer to evaluate a review as credible. In each discussion, we group the potential cues under major headings for organizational and readability purposes, rather than providing one long list of cues. It is important to understand that this research is exploratory, and we do not propose any particular factor structure to the cues.

3.4.1. Cues that a review is fake

Based on a thorough literature review, we identify five categories of aspects of online product reviews that potentially contribute to perceptions of the review as fake: Perceived Reviewer Inducement, Review Imbalance, Review Overlap, Review Composition Non-Conformity and Reviewer Implausibility.

Perceived Reviewer Inducement. After reading a review, consumers attempt to discern the motivation of the reviewer and then determine whether the reviewer has a stake in the customer's response (Clare et al., 2018; O'Reilly et al., 2016). Reviewers may have ulterior motives when they are compensated or may otherwise be perceived by customers as having a vested interest in brand sales.

Consumers are suspicious of reviewers who are compensated in some way, and for good reason. Paid reviewers are not often involved in the complete purchasing process, are less likely to follow up on product damage or delivery problems and are subject to positive bias (Dragen, 2016; Woolley & Sharif, 2021). These reviewers may have no experience with the product category and therefore be unable to reference competitors' offerings regarding how *relatively* good (or bad) the product is (Dragen, 2016; Woolley & Sharif, 2021). Thus, these reviews can provide inaccurate, biased or incomplete information about the brand that is then inconsistent with other reviews, causing confusion to potential customers. Furthermore, sponsored reviews harm brand perceptions. Stephen et al. (2012) demonstrate that brand evaluations (and subsequently willingness to pay for the product) are decreased when consumers are aware that reviewers are incentivized. DeVeirman & Hudders (2020) agree that incentivized reviews, once disclosed, produce skepticism, which, in turn, adversely affects brand perceptions. Interestingly, incentive disclosure by the reseller results in higher helpfulness ratings, but lower brand ratings and lower sales (Cui et al., 2022). Alternatively, when sponsored product reviews are not disclosed, consumers are unable to distinguish sponsored reviews from non-sponsored reviews (Pfeuffer et al., 2021).

In summary, sponsorship, inducement, or payment increases suspicion about the reviewer's ulterior motives (Kim, Maslowska & Tamaddoni, 2019) negatively affecting brand attitudes and purchasing. Similarly, the motives of reviewers are suspect as having a vested interest in brand sales when consumers are asked to buy, ‘click here’, or forward the review post (O'Reilly et al., 2016).

Review Imbalance. In a series of product reviews, it may be that there are more positive reviews than negative reviews. The ratio of positive to negative reviews can occur when a business either self-injects positive reviews (CMA 2015) or posts a series of negative reviews about business rivals. Fake positive reviews are more common than fake negative reviews, likely because the process is easier and less risky (CMA 2015).

Another means of affecting the review balance is by hiding or deleting negative reviews. While this does not involve posting false information, it does change the valence and distribution of reviews (Zhuang, Cui & Peng, 2018). One example is using various online tools to push negative reviews down the list where consumers are less likely to see them. According to Zhuang, Cui and Peng (2018), a smaller number (or total absence) of negative reviews may arouse customer suspicion. Regardless of the method, the greater presence of positive reviews over negative reviews increases consumer suspicion.

Furthermore, reviews may be extremely negative – or extremely positive. Compared to authentic reviews, fake reviews tend to be more extreme, given that firms are not motivated to post or support insipid or moderate reviews (Luca & Zervas, 2016). In either case, the extremeness of the reviews may arouse consumer suspicion.

Review Overlap. Patterns can emerge across a set of reviews that lead the consumer to suspect that reviews are fake. For example, when repetitive wording is detected across reviews, the consumer will not trust the reviews (Dragen, 2016). In fact, according to Joe Rohrich, Chief Revenue Officer for Bazaarvoice, the primary way customers identify reviews as fake is when multiple reviews use the same or similar wording (Schoolov, 2020). Furthermore, when multiple reviews are time-stamped very close to each other, even when the reviews appear to be posted by different reviewers, the reviews will be evaluated as fake (Dragen, 2016; see also Chatterjee et al. 2023). Reviews posted close in time suggest that there was a push for them to be posted on a strict timeline (Dragen, 2016).

Review Composition Nonconformity. Reviews that are easy to read and lack spelling and grammar errors should be, all else being equal, more influential compared to reviews that are difficult to read and have errors (Ghose & Ipeirotis, 2010). In a study conducted by Bazaarvoice, poor grammar and spelling errors were the third most common cue used by customers to evaluate an online review as fake (Schoolov, 2020).

Reviewer Implausibility. Many websites use checks to verify the authenticity of a user. One concern is whether the online review was generated by a bot. "Review tracking website, Fakespot, which analyzes reviews from popular e-commerce sites, estimates that a third of online reviews on sites like Walmart.com, Amazon.com, and Sephora.com are fake" (Khalid 2019).

Also of concern are situations where a real person is generating multiple fake user accounts to post the online reviews. For example, when reviewers use nonsensical usernames (such as a random combination of letters and numbers), it may indicate that the reviewer has generated multiple fake accounts. While a single product review using a nonsensical username may be considered fake, the presence of multiple reviews in the review set by other nonsensical names worsens the situation.

Another indication that the reviewer may not be authentic is when there is excessive use of clichés or hyperbole (e.g., "This is the last × you will ever have to buy!"). Fake reviews are generally expected to include more hyperbole than authentic reviews (Banerjee & Chua, 2017; Ott, Cardie & Hancock, 2013; Yoo & Gretzel, 2009) based on the 'conventional wisdom' that consumers should be suspicious of reviews that are too extreme to be true (Banerjee 2022). Banerjee (2022) finds that fake reviews are not necessarily more likely to include exaggeration, depending on whether the review is positive or negative, but this is not to say that when hyperbole is present consumers don't still use it as an indication of inauthenticity. Indeed, overstated reviews represent a type of fake review (Agnihotri and Bhattacharya, 2016; Baker and Kim, 2019; Harris, Fisk & Sysalova, 2016; Moon, Kim & Bergey, 2019). Additionally, phrases copied directly from the product's web page can suggest the review is fake.

3.4.2. Cues that a review is credible

Credibility (McCroskey & Teven, 1999) is associated with perceptions of objectivity, accuracy, and believability (Sundar, 1999). Online review credibility is more specifically defined as the extent to which a person perceives the review as believable, true, or factual (Park et al., 2011). Based on a thorough literature review, we identify the following four categories of aspects of an online review that potentially contribute to perceptions that a review is credible: Reviewer Validation, Review Detail, Purchase Validation, and Negative Reviews.

Reviewer Validation. When consumers consult product reviews to assist them in purchasing decisions, there may be cues that, if present, suggest that the reviewer him/herself is credible. Such cues include identification of the reviewer and various reviewer ratings or certifications.

Source credibility is particularly important in an online context because it is often difficult to know who the reviewer really is (Filieri et al., 2018). This is where reviewer identification comes in. Reviewer identification involves the use of the reviewer's real name as well as the

inclusion of the reviewer's photo, both of which enhance the perceived credibility of the reviewer (Baek, Ahn & Choi, 2012; Fogg et al., 2001; Liu & Park, 2015). Based on empirical research, Chatterjee et al. (2023) recommend that consumers know the identity of the reviewer before reaching a judgment regarding review credibility. Forman, Ghose & Wiesenfeld (2008) explain that consumers may use the personal information of the reviewer as a heuristic cue to make evaluations or take an action such as purchasing.

High reviewer rankings (in addition to reviewer identities) increase customer perceptions of review credibility (Baek, Ahn & Choi, 2012). That is, top-ranked reviewers who identify themselves with their real names come across as more credible and are therefore more trusted by consumers (Lee & Choeh, 2016). For example, Amazon.com provides rankings for reviewers who use their real names allowing consumers to know their identities and be in a position to evaluate their credibility (Baek, Ahn & Choi, 2012). Visitors to the site can see reviewer standings (e.g., "top-100 reviewer") and rankings, as well as personal information about the reviewer including the reviewer's real name, where they live, hobbies and interests, etc.

Review Detail. When it comes to the level of detail in a product review, more is better. The detail provided by product reviews may be thought of in terms of depth and specificity, length, internal balance, and updating. Depth and specificity are similar, yet different terms. Depth relates to the number of aspects or details included, while specificity refers to a lack of vagueness. Depth of information provides more product details, as well as details about how and where the product was used (Lee & Choeh, 2016). The depth of information in the review can help consumers distinguish among brands (Mudambi & Schuff, 2010), while specificity allows consumers to search for particular information regarding product attributes or features (Lee & Choeh 2016). In an empirical analysis of online reviews of movies, Kim, Park & Mariani (2023) find that credible reviews are associated with a more in-depth and specific analysis. Zhao et al. (2013) explain the greater the specificity, the more credible the information and the more influential the review.

The length or elaborateness of an online review also positively influences purchase intentions (Shelat & Egger 2002). That is, the longer and more elaborate the online review, the more likely consumers are to grasp the details regarding how and where the product was used (Lee & Choeh, 2016; Mudambi & Schuff, 2010; Racherla & Friske, 2012). Elaboration can prove to be powerfully persuasive (Liu & Park, 2015), reducing consumer uncertainty about product quality and increasing confidence in product evaluation and selection (Johnson & Payne, 1985).

Metzger (2007) agrees that long reviews that include vigorous explanations inspire consumer confidence in online purchasing. Compared to shorter reviews, longer reviews likely contain more information. Tversky and Kahneman (1974) suggest that the mere increase of information for a decision boosts consumer confidence. Research on consumer persuasion suggests as message processing increases, more favorable associations are formed (Petty & Cacioppo, 1984). Therefore, longer informative reviews that require greater mental processing may be perceived as more credible than shorter ones (Pan & Zhang, 2011).

Further, the length and the depth of a review may represent the reviewer's involvement in and knowledge of the product category. Involvement describes an individual's level of interest, search, or complex decision making (Zaichkowsky, 2010). The more involved reviewers are in the product category, the more information they are likely to have and share thorough comparisons to similar models and features. Highly involved reviewers are also more likely to be real users who offer trustworthy information to potential buyers (Pan & Zhang 2011).

Reviews that present both the pros and the cons of the product are considered internally balanced. Such two-sided reviews help consumers understand the favorable and unfavorable aspects of a product and tend to be more persuasive than one-sided reviews (Etgar & Goodwin, 1982; Kamins et al., 1989) in part because they provide information that goes

against the seller's interest (Kamins & Assael, 1987). Interestingly, Lutz, Pröllochs and Neumann (2022) note that the way in which the pros and cons are presented is important. More specifically, frequently alternating between positive and negative arguments serves to confuse customers based on information overload theory (Lutz, Pröllochs and Neumann, 2022). That said, alternating arguments does not contradict the claim that two-sided reviews are generally deemed more credible. In summary, two-sided reviews are generally perceived as more credible and trustworthy (Cheung et al., 2009; Filieri, 2016).

In some instances, a reviewer enters an update to his/her original post after some amount of time has passed. Additional detail is often provided in the update as the reviewer gains experience with the brand, lending greater review credibility.

Purchase Validation. Fake product reviews are often provided by reviewers with little or no experience with the products being reviewed (Zhang, Ko & Carpenter, 2016). Photos of the product in use provide evidence that the product has truly been purchased and used. Videos lend even greater support given that a video can show the product actively in use. The findings of Li et al. (2020) revealing that the majority of fake online reviews do not include photos further demonstrates that photos and videos of the product in use lend credibility to the online reviewer's experience with the product. In fact, based on empirical research, Chatterjee et al. (2023) recommend that consumers favor online reviews that are supported by photographs or v**ideos.

Purchases can be further validated by a 'verified buyer' badge (May 2011; Mayzlin, Dover & Chevalier, 2014) to indicate that the reviewer has indeed purchased the product (Lappas, Sabnis & Valkanas, 2016). The verified buyer badge is a popular tool adopted by multiple online sellers as a defense mechanism against fake reviews (May 2011; Mayzlin, Dover & Chevalier, 2014). Consumers use the verified buyer designation as an indication of the legitimacy of a product review. While many sites verify that the product was purchased (regardless of where it was purchased), others (such as Expedia) are more restrictive by only allowing guests who purchased on their site to provide reviews (Donaker, Kim and Luca, 2019).

Negative Product Reviews. Negative information is perceived as more credible than positive information (Kanouse, 1984). People place greater value on negative information, perceiving it to be more important and therefore allowing it to have a greater impact on their evaluation (Kanouse & Hanson, 1972). Accordingly, negative aspects may present as more believable than the positive contributions that support the seller's product claims.

In the next two sections, we present our two studies designed to address the two research questions. We begin with Study One to address the first research question regarding the consequences of suspicion that

a review is fake.

4. Study One: Consequences of fake product reviews

The first research question to be addressed is:

RQ1: What is the mechanism by which fake online reviews impact consumer purchasing decisions?

4.1. Research model and hypotheses

Our first study assesses the effect of suspicion that product reviews are fake on four potential consequences: (a) the value placed on reviewer's opinions, (b) consumer attitudes toward the brand and (c) the website, and (d) purchasing. We propose that a suspicion of product reviews as fake discounts the value of reviewer's opinion, which thereby reduces consumer attitudes toward the brand as well as the website, in turn decreasing the probability of completing the purchase (see Fig. 1 for conceptual model).

4.1.1. Customer suspicion and the value of the Reviewer's Opinion.

As discussed above, when consumers are presented with fake reviews, the immediate impact is on their perception of the reviewer. In other words, when consumers read a review, they immediately consider the motivation of the reviewer in preparing the review (O'Reilly et al., 2016). When consumers attribute the opinion presented in an online review to a persuasion goal, the nature of the attitude toward the reviewer is changed (Kim, Maslowska & Tamaddoni, 2019). In other words, consumers discount the value they place on the opinions and recommendations of the reviewer (Dellarocas, 2006; Mayzlin, 2006; Tuk et al., 2009). Rocklage and Fazio (2020) utilized a concept of 'emotionality backfires' to explain that readers usually enter the online review context with initially specific expectations, consciously or sub-consciously. Once reviewers' emotionality is at odds with consumers' expectations, consumers will be surprised and feel the reviews odd, with the consequence of discounting the review's opinions (Rocklage & Fazio, 2020). We therefore propose:

H1: Consumer suspicion of fake reviews will discount the value of reviewer opinion.

4.1.2. Value of the reviewers opinion and attitudes toward the brand and the website

A study by Kudeshia and Kumar (2016) empirically demonstrates that online word-of-mouth negatively impacts brand attitude. Fake reviews in particular arouse consumer distrust toward the product as well as the brand (Kim, Maslowska & Tamaddoni, 2019; Sparks & Browning,

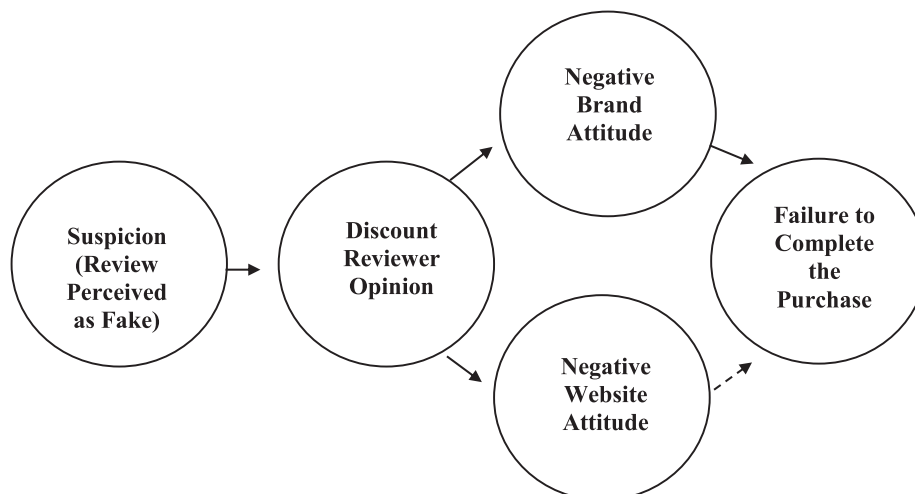


Fig. 1. Consequences of suspicion of online reviews as fake.

2011). This is consistent with the [Fake Online Reviews, 2021](#) report which finds that 26% of online buyers say the presence of fake online reviews causes them to distrust the brand.

We believe that it is the discounting of the reviewer's opinion which leads to a more negative brand attitude. As explained by [Campbell \(1995\)](#) and [Putz \(2017\)](#), when the reviewer's motives are perceived as extrinsic, consumers will react to the reviewer and negatively evaluate the brand as a result. [Lin and Xu \(2017\)](#) empirically demonstrate that the consumer's ability to trust the reviewer directly affects attitude toward the brand. Similarly, [Verma et al. \(2023\)](#) find that online opinion adoption (somewhat the opposite of discounting the reviewer opinion) has a positive effect on attitude toward the brand. More directly relevant to the effects of fake reviews, [Karabas et al. \(2020\)](#) report that consumers' perceptions of manipulative intent by the reviewer lead to more negative brand attitudes. Although manufacturers may benefit in the short term from manipulated reviews, once the reviews are deemed fake, consumer trust toward the overall product reviews may be lost, producing a detrimental effect on attitudes toward the brand ([Krishnan & Wan, 2021](#)). We therefore suggest that the mechanism by which suspicion affects brand attitude is through the discounting of the reviewer's opinion when suspicion is aroused that the review is fake. Our hypothesis is as follows:

H2: Discounting of the reviewer's opinion will adversely change the attitude towards the brand.

As noted by [Kim, Maslowska & Tamaddoni \(2019\)](#), distrust toward a reviewer may lead the customer to distrust not only the brand but also the website. However, there appears to be little to no empirical research to date regarding discounting the reviewer's opinion on attitude toward the website. That said, [Dwivedi et al. \(2021\)](#) find that manipulation of online reviews does indeed influence consumer attitudes toward the online platform. Accordingly, we argue that when information on a company's website (including product reviews) cannot be relied upon by customers, consumer trust in the website is eroded. Recent research by [BrightLocal, 2022](#) found that fake reviews on popular sites like Google, Amazon and Facebook significantly damages trust in those companies and the reviews they host. It is therefore our contention that:

H3: Discounting of the reviewer's opinion will adversely change the attitude towards the website.

4.1.3. Attitudes toward the brand and Completing the purchase

An important aspect of a brand is its implied promise and trustworthiness to deliver something of value to consumers ([Rodgers & Thorson, 2012](#)). It is well known that brands make promises and try to create trust between the producer and consumer ([Rodgers & Thorson, 2012](#)). Establishing trust is important for any firm selling its products online ([Cheng & Loi, 2014](#); [Rodgers & Thorson, 2012](#); [Sparks & Browning, 2011](#)). In fact, [Dash, Kiefer and Paul \(2021\)](#) recently established that brand identity and image are the most two significant factors influencing purchase intention. When the trust and belief in the brand is broken due to the false information included in fake reviews, the product evaluation and decision-making process is inhibited. When consumers realize that information in the review is not credible, they are left without valid information on which to base a decision. In summary, negative brand attitudes lead to reduced purchase intentions ([Karabas et al., 2020](#); [Kudeshia and Kumar, 2016](#); see also [Verma et al. 2023](#)).

H4: Negative attitude toward the brand will decrease consumers' completion of the purchase.

4.1.4. Attitudes toward the website and Completing the purchase

Existing research regarding the effect of a negative attitude toward the website on purchasing is unclear. One school of thought, as mentioned previously, argues that because consumers know fake reviews exist and are sufficiently able to adjust their interpretations ([Zhuang, Cui & Peng, 2018](#)), fake reviews may *not* impact consumer perceptions of the website. It is also possible that even if customers hold negative attitudes toward the website based on perceptions of fake

reviews, they may purchase the brand anyway for other reasons, such as a better price. Alternatively, when consumers distrust the reviewer and the review, the negative sentiment will spill over not only to the brand but also to the website ([Kim, Maslowska & Tamaddoni, 2019](#); [Roman, Riquelme, & Iacobucci 2023](#); [Sparks & Browning, 2011](#)). This can be explained by the theory of the transfer of trust ([Roman, Riquelme, & Iacobucci 2023](#)).

[Köksal & Penez \(2015\)](#) contend that if the website does not lead the consumer to believe that the merchant is trustworthy, no purchase decision will result. [Sullivan and Kim \(2018\)](#) suggest that consumers depend on perceptions of online trust to determine future purchasing. Although findings are mixed, we tentatively believe that having a negative attitude toward the website will have a detrimental effect on purchasing. A distrust of the website arising from suspicion of fake online reviews may be sufficient cause for consumers to abandon the transaction. We therefore propose:

H5: Negative attitude toward the website will decrease consumers' completion of the purchase.

4.2. Study One: Methodology

For the current study, participants were asked to recall a recent purchasing experience of an electronics product (laptop, cell phone, smart watch, Bluetooth headphones, etc.) where they used online reviews in their information search. They were encouraged to recall as much as they could about the experience before responding to the survey questions. Consumer electronics is an appropriate product category for which to examine the role of online reviews in consumer purchasing. According to the [Fake Online Reviews 2021](#) report, of the \$95 billion in online electronics sales, \$84.5 billion are affected by online reviews and \$3.3 billion in consumer spending is influenced by fake online reviews. The global value of consumer electronics is expected to reach \$1,787 billion in 2024, with online sales suggested to account for almost half (47 percent) by 2023 ([Bayard, 2015](#)). Furthermore, electronics products are considered to be experience goods, where the evaluation of product quality depends on experience with the product as opposed to objective search attributes. Subjective online reviews are more important in the decision-making process for experience goods ([Mudambi & Schuff, 2010](#); [Pan & Zhang, 2011](#)).

4.2.1. Data collection

The sample was comprised of 222 responses from a student panel at a southwestern university, 71.6% of which were female. In terms of age, 44% were between the ages of 25 and 34, but the sample included respondents over the age of 44 (7%). Although the appropriateness of student samples in marketing research has been debated for years ([Chen, Schwartz & Yu, 2015](#)), such use may be supported by (a) acceptance in high level research and (b) justification of students as a relevant population given the research topic ([Harrison-Walker, 2019](#)). In a survey of over 500 students, 60 percent report spending money monthly on electronics ([College Students' Spending Habits, 2023](#)). The target population of interest for the current research consists of consumers experienced with purchasing electronics products online and therefore a student population is appropriate. (Note that we will retest and therefore validate the model using a more generalized sample of respondents in Study Two.).

4.2.2. Measures

For our perception of an online product review as fake (e.g., Suspicion), we adapted the nine-item Skepticism scale by [Zhang, Ko & Carpenter \(2016\)](#). Measures for the consequences investigated in the first study (Discounting Reviewer Opinion, Negative Brand Attitude, Negative Attitude Toward the Website, and Not Completing a Purchase) were developed based on existing literature. Items used to assess these consequences are presented in Appendix A.

4.2.3. Data analysis

We began by assessing the scales developed for the four outcome measures. Each scale was purified employing an iterative process of evaluating item-remainder correlations, internal consistency, and construct dimensionality according to the recommendations set forth by Churchill (1979) and Spector (1992). The next step was to apply Bartlett's Test for Sphericity (BTS) and Kaiser's Measure of Sampling Adequacy (KMSA) to each scale. When the BTS statistic is significant, it indicates that the data is appropriate for factor analysis. With regard to KMSA, an assessment of how small the partial correlations are relative to the ordinary correlations, values greater than 0.8 can be considered good (SAS User's Guide, p.798). Finally, factor analysis was conducted using IBM SPSS Statistics 22 software to assess dimensionality. Each of the four outcome constructs were found to be unidimensional.

Discounting Reviewer Opinion. All five items were retained following scale purification, yielding a reliability of 0.902. BTS suggests that the bivariate correlations among the scale items were significantly different from zero (650.242, 10 df, sig. = 0.000), while KMSA was evaluated as good (0.863).

Negative Brand Attitude. The Negative Brand Attitude scale consisted of five items. Using the iterative process described above, one item (ATB5) was deleted. The resulting reliability for the four-item scale was 0.923. BTS was significant (615.282, 6 df, sig. = 0.000) and the KMSA was good (0.840).

Negative Attitude Toward the Website. All five items were retained following scale purification, leading to a reliability of 0.894. BTS was significant (620.813, 10 df, sig. = 0.000) and the KMSA was good (0.837).

Not Completing a Purchase. This scale consisted of five items. However, one item (PUR3) was removed using the iterative process of scale purification described above. Scale reliability for the remaining four items was 0.923. BTS was significant (631.547, 6 df, sig. = 0.000) while the KMSA was good (0.852).

Suspicion. We then conduct factor analysis on the nine Suspicion items to assess dimensionality. Contrary to the findings of Zhang, Ko & Carpenter (2016) who reported three factors of Truthfulness, Motivation and Identity, the adapted nine-item scale used in the current research was found to be unidimensional with 59.8% of the variance explained. Reliability for the nine-item Suspicion scale was 0.916. Further, both the BTS (1068.385, 36 df, sig. = 0.000) and the KMSA (0.919) were acceptable.

4.3. Results

The causal relationships were tested using LISREL 8.80 adopting the recommendations by Bagozzi and Heatherton (1994) using composite variables. When there are four or more items per construct, the model is unlikely to fit (Bagozzi & Heatherton, 1994). Accordingly, for each of the four outcome constructs, we divided the items to create two composite variables. For the Suspicion construct, we used the domain parceling method (Kishton & Widaman, 1994) which is justified when using structural equation modeling because "SEM makes the assumption that items are drawn from the same domain" (Graham, Tatterson & Widaman, 2000). Before considering the structural links, we examined the model fit. While the chi square value is significant (Chi square = 89.57, df = 40, $p < 0.00001$), the ratio of chi square to degrees of freedom is 2.24. Although there is no consensus regarding an acceptable ratio, recommendations vary from 2.0 (Tabachnick & Fidell, 2007) to 5.0 (Wheaton et al., 1977). The Goodness of Fit index (0.93), Adjusted GFI (0.89), Comparative Fit Index (0.99), Standardized Root Mean Square Residual Index (0.024) and the Root Mean Square Error of Approximation (0.075) support acceptable model fit.

4.3.1. Hypothesis testing

H1 predicts that Suspicion will have a positive direct effect on Discounting Reviewer Opinion. This hypothesis is supported ($p = 0.000$).

H2 and H3, respectively, propose that Discounting Reviewer Opinion will positively affect Negative Attitude toward the Brand and Negative Attitude toward the Website. Both H2 and H3 are supported ($p = 0.000$). H4 suggests that Negative Attitude toward the Brand has a positive effect on Failing to Complete the Purchase. This hypothesis is also supported ($p = 0.000$). Finally, H5 suggests that Negative Attitude toward the Website has a positive effect on Failing to Complete the Purchase, however this hypothesis is not supported ($p = 0.10$). The results of the structural equation model are presented in Table 1. Discussion of the results will be presented following Study Two, which is our next section.

5. Study two: Cues used by consumers to evaluate reviews

Now that the significance of suspicion of fake online reviews is established and we have reached a better understanding of the consequences on consumer purchasing, we move on to investigate our second research question:

RQ2: *What are the cues that consumers use to evaluate online product reviews as fake, (versus reviews that are credible)?*

5.1. Methodology

In Study Two, we empirically determine which of the factors identified in the literature review actually impact consumer evaluation of an online product review as fake (or credible). We also explore the factor structure for each set of cues to gain a better understanding of how the cues may be organized in consumers' minds.

5.1.1. Face Validation

Before launching a national online survey and given that the research regarding cues is exploratory, we wanted to pretest the 33 items (see Appendix B) by asking respondents to indicate for each item whether it (a) signals that a review is credible, (b) signals a review is fake, or (c) has no impact on the evaluation of a review. The goal in part was to weed out items from the full list, if any, that consumers do not employ when evaluating the veracity of online reviews.

5.1.2. Sample for Pre-Testing

The pre-test sample was comprised of 40 students, ages ranging 19–47 with an average age of 30.3, 70% female, and a ratio of 3:1 graduate versus undergraduate level status. Items were retained for the study if at least 50 percent of the respondents categorized the item as either credible or fake. Of the 33 items total, thirteen were not supported while 20 received support. The three items of extremely positive reviews, extremely negative reviews, and the ratio of positive to negative reviews were deleted (which means all of the items originally listed

Table 1
Study One: SEM Results for the Effects of Suspicion.

Direct Effect	Estimate	Standard Error	T-Value
Suspicion	0.82	(0.07)	12.21*
→ Discount Reviewer Opinion			
Discount Reviewer Opinion	0.91	(0.05)	16.61*
→ Negative Attitude toward the Brand			
Discount Reviewer Opinion	0.97	(0.05)	18.30*
→ Negative Attitude toward the Website			
Negative Attitude toward the Brand	0.93	(0.05)	19.93*
→ Failure to Purchase			
Negative Attitude toward the Website	−0.13	(0.08)	−1.52
→ Failure to Purchase			

* $p = 0.000$.

under Review Imbalance). The two items intended to address concerns with grammar and spelling (Review Composition Nonconformity) as well as the two items regarding unfavorable reviews (Negative Reviews) and therefore their corresponding categories were also omitted from the study. The remaining items that were not strongly classified as either signals of credibility or inauthenticity included IND1 (reviewers receiving discounts for the review), OVR2 (multiple reviews that sound similar), RVL2 (photo of the reviewer), and DET3 (very long reviews). These individual items were removed from the study. Accordingly, six of the original nine categories and 20 of the individual measures were supported by the pretest.

5.2. Results

5.2.1. Consequences of suspicion model Validation

To retest our conceptual model from Study One, we collected data from a national online panel of U.S. respondents who were paid a small remuneration for their participation. The quality of the responses was supported using Cloud Research Approved Participants who had completed at least 50 projects in the past and earned an approval rating of 60% or greater. In the study description, prospective respondents were advised: "To respond to this survey, you must have recently purchased an electronics good online (such as cell phones, laptop computers/tablets, or smart watches) and used online product reviews in making your final purchase decision." Out of 250 total responses, 237 were found to be useable. The three reasons for rejecting responses were cases that clearly stated no online reviews were utilized (even though that requirement was clearly stated), copying and pasting filler words (such as copying the question itself) when asked to describe a specific purchasing experience, and respondents that based their answers on the purchase of a service rather than a physical, electronics good.

Based on 237 responses, 62.9% were from female participants. The age range of the respondents was from 19 to 79 years, with a mean age of 41.6 years. In terms of the highest level of education completed, 39.2% held a bachelor's degree, 15.6% held a graduate degree and 32.1% either had some college or earned an associate's degree.

We first re-examine the four potential consequences of Suspicion that the online reviews were fake using the same process that was used in Study One, this time omitting the insignificant path from Negative Attitude Toward the Website to Failure to Purchase. The results are shown in Table 2. While the chi square value is significant (Chi square = 129.64, $df = 40$, $p < 0.00001$), the ratio of chi square to degrees of freedom is 3.241. The Goodness of Fit index (0.908), Adjusted GFI (0.847), Comparative Fit Index (0.976), Standardized Root Mean Square Residual Index (0.0246) and the Root Mean Square Error of Approximation (0.0972) support acceptable model fit. The findings from the online panel are consistent with those from Study One in that H1 through H4 are supported.

Table 2
Study Two: SEM Results for the Effects of Suspicion.

Direct Effect	Estimate	Standard Error	T-Value
Suspicion → Discount Reviewer Opinion	0.86	(0.06)	15.48*
Discount Reviewer Opinion → Negative Attitude toward the Brand	0.94	(0.05)	21.08*
Negative Attitude toward the Brand → Discount Reviewer Opinion	1.024	(0.04)	26.30*
Negative Attitude toward the Website → Negative Attitude toward the Brand	1.01	(0.03)	34.97*
Negative Attitude toward the Brand → Failure to Purchase			

* $p = 0.000$.

5.2.2. Cues Validation retest

To test the validity for our items representing cues, we again asked respondents to consider each item (see Appendix B) and indicate whether the item (a) signals a review is credible, (b) signals a review is fake, or (c) has no impact. The findings were highly consistent with those of the student sample, with only two exceptions. Two items (IND1 and OVR2) were not supported in the student sample but were identified as signaling that a review is fake in the online sample. Thus, these two items were maintained for factor analysis. Of the 22 measures retained, 11 were associated with fake reviews and 11 were associated with credible reviews based on the sorting process.

5.2.3. Factor analysis

The next step was to explore the dimensionality of the supported cues for fake reviews. Respondents were asked to what extent they agreed that the item listed, if present, indicates a product review is inauthentic (or fake). A 7-point Likert-type scale ranging from strongly disagree to strongly agree was used. The results are presented in Table 3. The findings indicate that cues that an online product review is fake may be conceptualized as having two factors. The first factor we call Implausibility, and this includes the items we initially associated with reviewer implausibility, the items associated with review overlap (bringing the review into question), and item IND4 which asks a viewer to buy, click or forward the review (which could be about the reviewer or the review). Thus, the first factor of Implausibility includes 8 items. The second factor is Inducement, comprised of three items (IND1-IND3). Together, the two factors explain 57.8% of the variance. Table 3 presents the scale reliabilities for the two factors.

Finally, we consider the factor structure of the supported cues for credible reviews. Respondents were asked to what extent they agreed that the item listed, if present, indicates a product review is authentic (or credible). Results suggest that authenticity cues comprise two factors. Review Detail includes five of the original detail-related items as well as the two items mentioning the inclusion of a picture or video of the product. It seems that respondents feel the picture and video add detail to the product review. The second factor, Validation, includes a purchase validation related item, two reviewer validation related items, and one detail related item. Respondents indicate that a balanced view of a product, including pros and cons, validates the review. Crowley & Hoyer (1994) explain that two-sided messages increase consumer perceptions that the opinion is objective. These findings are consistent with the meta-analysis conducted by Verma et al. (2023) supporting that both review and reviewer aspects are involved in evaluations of credibility. Together, the two factors explain 58.0% of the variance. Scale reliabilities are reported in Table 4.

In summary, our research was able to identify a set of cues that customers use to indicate an online review is fake and a second set of cues used to evaluate an online review as credible. Each scale was comprised of eleven items, two factors each, with demonstrated

Table 3
Study Two: Factor Analysis of Cues that Indicate a Review is Fake.

Item	Factor 1: Implausibility	Factor 2: Inducement
	Chronbach's Alpha = 0.860	Chronbach's Alpha = 0.736
IMP1	0.853	
IMP2	0.823	
OVR3	0.683	
OVR2	0.620	
IMP3	0.602	
IND4	0.567	
IMP4	0.555	
OVR1	0.524	
IND2		0.824
IND1		0.775
IND3		0.700

Bartlett's Test of Sphericity: Chi Square = 1063.385, Sig. = 0.000.
KMO measure of Sampling Adequacy = 0.859.

Table 4

Study Two: Factor Analysis of Cues that Indicate a Review is Credible.

Item	Factor 1: Review Detail Chronbach's Alpha = 0.881	Factor 2: Validation Chronbach's Alpha = 0.721
DET2	0.853	
DET1	0.841	
DET4	0.667	
DET7	0.630	
PVL1	0.620	
PVL2	0.605	
DET6	0.570	
PVL3		0.776
RVL4		0.677
RVL1		0.668
DET5		0.591

Bartlett's Test of Sphericity: Chi Square = 1251.155, Sig. = 0.000.

KMO measure of Sampling Adequacy = 0.867.

reliability.

6. Discussion of results

In this article, we raise two critical research questions. First, what are the attitudinal and behavioral consequences with regard to online purchasing when consumers are suspicious of online reviews and what are the relationships between and among the constructs? This first question establishes the importance of investigating the effects of fake online reviews. Second, what are the cues that consumers use to evaluate online product reviews as fake (versus credible)?

Our discussion begins by first examining the findings related to the four consequences of Discounting Reviewer Opinion, Negative Attitude toward the Brand, Negative Attitude toward the Website, and Failure to Complete a Purchase. The scales utilized to measure these constructs were found to be highly reliable. As expected, the reviews that are perceived as fake reduce the value of the reviewer's opinion. This support for H1 is consistent with previous suggestions that when consumers suspect a review is fake, they being to discount the reviewer's opinion (Kim, Maslowska & Tamaddoni, 2019; O'Reilly et al., 2016; Rocklage & Fazio, 2020; Tuk et al., 2009).

H2, which states that devaluing the reviewer's opinion leads to negative attitudes towards the brand, is supported by the current research. This finding is consistent with assertions from Campbell (1995), Karabas et al. (2020), Lin and Xu (2017), and Putz (2017).

H3, also supported, expects that discounting of the reviewer's opinion will be associated with a negative attitude toward the website. H3 received literature support from Kim, Maslowska, & Tamaddoni, 2019; BrightLocal, 2022.

Consistent with our hypothesis 4, a negative attitude toward the brand interferes with the completion of the purchase. H4 was based on arguments by several researchers (Kim, Maslowska & Tamaddoni, 2019; Koksál & Penez, 2015; Sparks & Browning, 2011; Sullivan & Kim, 2018).

Contrary to our fifth hypothesis (H5), however, a negative attitude toward the website does not affect the completion of the purchase. Recall that the literature support for H5 was unclear. Koksál & Penez (2015) and Sullivan and Kim (2018) provide support for H5 explaining website distrust interferes with consumer purchasing, while Zhuang, Cui & Peng (2018) argue that consumers know about fake reviews and adjust their interpretations accordingly. More recently, Roman, Riquelme & Iacobucci (2023) present and test a model showing that while review authenticity affects attitude toward the website, they find no path between attitude toward the website and purchase intentions. No explanation is offered. The non-significant finding of a relationship between attitude toward the website and completion of purchase deserves further attention.

The unexpected insignificant relationship between the Negative Attitude toward the Website and the Failure to Complete a Purchase can be explained by one of the findings of O'Neil and Eisenmann (2017) that

consumers tend to consult a number of sources including different types of websites for product information. According to O'Neil and Eisenmann (2017), the lines between media sources are blurring and as a result, consumers may not recall the website where they found the information. People are more concerned with the quality of the information than about the website presenting the information (O'Neil & Eisenmann, 2017). Once consumers become suspicious of the online reviews on a certain website, they are likely to switch to other sources. If consumers succeed in finding the information they trust, they will complete a purchase. In other words, customers who form a negative attitude toward the website may still purchase the brand, but from a different website. Alternatively, the customer may form a negative opinion of the website but purchase the brand from that website anyway for other reasons, such as having the best price, free shipping, fast delivery, in stock availability, or the site being the only place the brand can be purchased. Another explanation is the strategy of retargeting by the website; that is, when a consumer abandons items in the online shopping cart, the seller may reach out with personalized encouragement to buy and offering price incentives (Keller and Swaminathan, 2020). Thus, even with a negative attitude, the purchase may be completed.

In summary, the perception of an online review as fake leads consumers to discount the reviewer's opinion, which in turn has a negative effect on the consumer's attitude toward the brand driving down product sales. The findings regarding the negative consequences of consumer suspicion that a review is fake establish the criticality of better understanding what exactly leads consumers to suspect an online review is fake.

This brings us to the second research question and the attempt to identify cues that signal to a consumer that the review is fake (or credible). We began with a total of 33 potential cues that were subjected to sorting for validation by two separate panels followed by factor analysis. This produced a list of 11 items indicating a review is fake, 11 items indicating a review is credible, and 11 items that had no impact. Examples of cues that consumers use to suspect that a review is fake are reviews posted by reviewers with nonsensical usernames, multiple reviews of the same product using non-sensical names, and reviews that disclose that the reviewer has been paid or compensated for the review. Cues that consumers use to evaluate a review as credible include reviews that are specific in detail, reviews that are very in-depth, and reviews that provide verified buyer badges. The results of the factor analysis indicated two factors of cues indicating a review is fake (Inducement and Implausibility) and two factors of cues indicating a review is credible (Review Detail and Validation). While the factor structure may be useful in future empirical research (note the factor structure needs to be confirmed), the individual items indicating a review is fake and those indicating a review is credible are more valuable to marketing managers. The 11 items indicating a review is fake and the 11 items indicating a review is credible are identified in Appendix B.

In addition to pinpointing 22 items that are used by consumers in evaluating the authenticity of online reviews, we also identify 11 items that consumers deem unimportant in making this evaluation. More specifically, reviews by experts with a vested interest (Clare et al. 2018; O'Reilly et al., 2016), extremely positive or negative reviews (Luca & Zervas, 2016), more positive than negative reviews (Zhuang, Cui and Peng 2018), and reviews with spelling or grammatical errors (Ghose & Ipeiritos, 2011; Schoolov, 2020) are suggested in prior literature as indicators of fake reviews, but are not supported by the current research. In hindsight, the first item may have been confusing to respondents since it included potentially conflicting aspects of 'expert reviewer' (which might infer credibility) and 'vested interest' (which might suggest inauthenticity). It appears that valence and review balance may not by themselves be useful since brands themselves vary in quality and could be associated with both good and bad reviews. For example, a very low-quality product would yield a large number of negative reviews, many more than positive ones. Finally, reviews with spelling or grammatical errors may be interpreted as simply the way average people talk, and

therefore unrelated to the authenticity of the message.

The remaining five cues that were found to be non-significant in our research include *reviewer* photos (Baek, Ahn & Choi, 2012; Fogg et al., 2001; Liu & Park, 2015), very long reviews (Tversky and Kahneman 1974), negative and unfavorable reviews (Kanouse, 1984; Kanouse & Hanson, 1972), and reviewer ratings (Baek, Ahn & Choi, 2012). These cues were originally proposed by existing literature to support that a review is credible. It is possible that the effect of reviewer photos could depend on the personal reaction of the consumer to the individual reviewer's photo. Long reviews in and of themselves did not suggest credibility and this likely relates to the exact content rather than the length of the review. A negative review could be associated with a low-quality product, or it could be posted by someone with a vested interest in seeing the product fail, such as a competitor. Therefore, the valence in and of itself may not be useful in evaluating authenticity. Finally, the effect of *reviewer* ratings on evaluating review credibility may depend on how such ratings are determined. For example, Amazon bases its reviewer rankings on the number of reviews submitted, the recency of the reviews and the quality of reviews based on 'yes' votes by consumers (Wong, n.d.). The number and recency of reviews are unrelated to authenticity, while the 'quality' of reviews may be based on criteria other than authenticity. For example, a vote of 'no' may indicate a review is not helpful, which is not to say that the review is fake.

In summary, the current research offers at least six distinct contributions. First, we demonstrate the mechanism by which suspicion that a review is fake indirectly affects purchase completion, identifying the missing link of discounting the reviewer's opinion. When consumers see the review information as unreliable, they deem the *reviewer* not credible (see Gössling, Hall and Andersson, 2016). Second, we report the unanticipated finding that brand attitude affects purchase completion, but attitude toward the website does not. We presented potential reasons for this finding and identify this as an area that could be explored in more detail in future research below. Third, we establish that the cues used by consumers to evaluate a review as fake are not the same cues as those used by consumers as credible. Fourth, we use exploratory factor analysis to suggest that each set of cues is comprised of two factors. The two factors indicating a review is fake are referred to as Inducement and Implausibility, while the two factors indicating a review is credible are termed Review Detail and Validation. Fifth, we arrive at the conclusion that there are 11 items indicating a review is fake and 11 different items indicating a review is authentic by asking consumers directly which cues they use in each case. Finally, we identify eleven cues originally suggested in previous literature as impactful that consumers indicate are not actually used in their evaluation of a review as either fake or credible. The findings from both Study One and Study Two suggest several implications for managers as well as for future research.

7. Contributions, Implications, limitations and future research

In the following sections, we present the (1) theoretical contributions, (2) practical and managerial implications, and (3) limitations and future research.

7.1. Theoretical contributions

Our research makes a number of contributions to the literature. First, we contribute to the literature on persuasion knowledge (Friestad & Wright, 1994) by investigating how consumers evaluate persuasion attempts in the form of online reviews that are designed to affect consumer attitudes and behaviors toward a purchase. Briefly, consumers develop an awareness of the tactics used in marketing communications intended to influence their attitudes and behaviors (Friestad & Wright, 1994). Over time, they learn how to adaptively respond to such attempts in a way that helps them achieve their own goals (Friestad & Wright, 1994). The consumer's persuasion-related knowledge structure is activated (Fein, Hilton & Miller, 1990) when the consumer becomes suspicious

that the review may not be legitimate, causing the consumer to question both the reviewer and the review (DeCarlo, 2005). This study empirically demonstrates that when consumers become suspicious about an online review, their opinion of the reviewer is reduced and therefore they are less persuaded. We demonstrate that suspicion activates the persuasion knowledge structure. In other words, the suspicion does not have a direct effect on attitudes toward the brand, attitudes toward the website or purchasing intentions (which Friestad & Wright 1994 may refer to as persuasion coping), but rather suspicion directly affects the persuasion knowledge structure.

Second, the current research contributes to the literature on referral congruity. When consumers read an online review and perceive an incongruity, they engage in greater information processing (Lee & Schumann, 2004), become suspicious of the authenticity of the review (Putz 2017), and question the reviewer's underlying motives (Pyszczynski & Greenberg, 1981) for publishing the review. Incongruity in our research refers to a mismatch between the online review and the knowledge structure that serves as a frame of reference for arriving at an evaluation. As explained by Lee and Schumann (2004), consumers reject information that lies beyond the range of acceptability. We have extended the theory presented by Lee & Schumann (2004) who adopted the theory of incongruity to advertising by applying the theory to online reviews. This study has empirically demonstrated that the incongruity between information derived from the online review and the frame of reference based on a normal range of expectations leads to greater information processing, the evaluation of the review as fake and suspicion of the reviewer's motives. More specifically, consumers use several cues from the online review that suggest implausibility (e.g., nonsensical usernames, the excessive use of superlatives, copied and repeated text, etc.) because such things lie beyond the range of typical expectations for a review. In other words, these cues jump out to consumers, making them think more carefully about what they are reading, and lead to suspicion that the review is fake.

Third, this research contributes to the literature on motive attribution (see Putz 2017). We apply the theory of motive attribution to consumer evaluation of online reviews. Specifically, consumers often perceive extrinsically motivated reviews as associated with some form of incentive and therefore less credible. The current research lends empirical support for the role of motive attribution finding that perceived inducements (e.g., extrinsic motivation) to post a fake review are in fact one of the two cue factors used by consumers to evaluate a review as fake. More specifically, the current research finds that inducements for fake reviews include when the reviewer receives a product discount, compensation, or entry into a sweepstakes. When inducements such as these are present, consumers find the reviews to be less credible.

Finally, the current research demonstrates that the evaluation of reviews as inauthentic is not the opposite of the evaluation of reviews as authentic. This is important to theoretical development and consequent empirical testing. Consumer's use cues to evaluate an online product review. The cues used to identify an online review as inauthentic are not the same as cues used to evaluate that an online review is credible. This is important for future research in that measures based on a given set of cues of inauthenticity cannot simply be reverse-scored to arrive at a measure of authenticity. Conceptual models designed for future research need to investigate cues for authenticity and cues for inauthenticity separately.

7.2. Practical and managerial implications

Our first research question relates to the mechanism by which fake online reviews impact consumer purchasing decisions. Our findings suggest that suspicion that a review is fake first leads to discounting the reviewer's opinion, which in turn negatively affects attitudes toward the brand and the website. In other words, suspicion first impacts how the consumer perceives the *reviewer*. As explained by Gössling, Hall &

Andersson (2016), there is a difference between trust (or suspicion) of the online information presented and the perceived credibility of the online reviewer. As marketers, we need the reviewer's opinions to be respected and valued so that neither our brand nor our website is harmed. Even if the review of our brand is negative (e.g., the brand is not satisfying customers), as long as the reviewer is perceived as legitimate, we have the opportunity to respond and make things right, which can then be communicated to consumers in the form of an updated review. In fact, the current research finds that updated reviews tend to support consumer perceptions of authenticity.

Means by which the reviewer's legitimacy can be supported depend to some extent on the role of the marketer as either the brand manager or the online reseller, and as we see from the findings, both roles are important. Because the reviewer's discounted opinion negatively affects the brand as well as the reseller's website, both the manufacturer and the reseller have a vested interest in reducing or eliminating fake online reviews to avoid damage to the brand and the website. Managerial implications for the manufacturer are limited by what the brand manager can control or significantly influence. Most branded products are sold via third party websites that are not under the manufacturer's control. Armed with the findings from this research, the manufacturer should attempt to work with sites that are authorized resellers to improve the quality of product reviews with specific focus on the cues identified in this study, relating that consumer attitudes toward the seller's website may also be negatively affected by reviews deemed inauthentic by consumers. For example, resellers should be discouraged from offering inducements in exchange for positive reviews since these reviewers may have limited involvement with the brand or product category, may not experience the complete purchasing process (including post-purchase problems with the brand and failing to provide review updates), and may be positively biased (Dragen, 2016; Wooley and Sharif, 2021). This recommendation goes against the current trend of incentivizing individuals to post reviews about a brand (Abendroth & Heyman, 2013; He, Hollenbeck & Proserpio, 2021). An alternative strategy that is rising in popularity is for brand managers to compensate key influencers and celebrity endorsers to provide opinions about the brand. "Because followers trust influencers as experts in their industries, marketers can collaborate with influencers to promote their brands through recommendations or featured content" (Fallon and Medina, 2021). For example, Mark Fischbach is a key influencer in the gaming and entertainment industry, while consumers look to Kayla Itsines for recommendations regarding health and fitness (Fallon & Medina 2021).

Further, manufacturers can be proactive with the third-party website by (1) establishing a specific manager to track review fraud and then (2) reporting fraudulent reviews to the site hosting the reviews (Sussin & Thompson, 2013) in a cooperative manner for the benefit of both parties. For example, Google allows individuals to flag a review, which then links to the Report a Policy Violation page to report the review as fake (How to Deal with Fake Online Reviews of Your Business 2020). By comparison, Amazon allows individuals to click on the Report Abuse button below a review; however, Amazon no longer allows one to comment on the reason the report is being made (Middleton, 2021). Manufacturers and resellers need to work together to remove fake online reviews promptly since both the brand and the seller's website are harmed by the presence of these reviews. It is not only the fake review that should be removed, but also the fraudulent reviewer, much in the same way that individuals posting dis- and mis-information on social media sites are banned from the platforms. The brand manager and the reseller could publish press releases touting their success in removing fake reviews and fake reviewers, thus providing greater comfort to prospective customers who rely on online reviews for decision-making.

The second question addressed by the current research was to identify the cues that consumers use to evaluate online product reviews as fake (versus reviews that are credible). We have already addressed the concern regarding compensation of reviewers that may be more under the control of the brand manager, so we turn now to strategies indicated

for the third-party reseller. It seems that these website managers are in better position to have a direct impact on reducing or eliminating fake reviews posted by suspect reviewers. For example, most reseller websites use artificial intelligence review systems to detect patterns in phrasing by single reviewers and across reviewers to identify fake reviews. Based on the findings in the current research that multiple reviews with similar or repetitive wording and reviews that are time-stamped close together serve as cues that the reviews are fake, it is critical to eliminate these reviews from the website. Similarly, reviews that copy direct phrasing from the brand's website can be detected by AI, while the reviewer's use of clichés or hyperbole or written by reviewers with nonsensical usernames warrant greater scrutiny. The use of verified user badges could be helpful in discriminating nonsensical usernames. In any case, the more data that is collected, the more the IA engine can adapt and learn (Dragen, 2016). It's essential that websites remove the fake posts immediately upon detection. According to findings reported by He, Hollenbeck and Proserpio (2020), Amazon was deleting about 40% of fake reviews in 2019 but was taking an average of 100 days to remove the posts. In that period of time, both the brand and website perceptions have been damaged.

A significant contribution of this study is differentiating cues that consumers use to conclude that a post is fake versus cues used to consider a post authentic. Importantly, a number of cues identified in the literature as impacting consumer evaluations were demonstrated in this study not to be used by consumers in real-life (as opposed to findings from scenario or experimental research) decision making. For example, some have suggested that spelling and grammar errors serve as cues that an online review is fake (Schoolov, 2020). Our research shows that such errors do not have an overall significant effect. It is reasonable that in some cases spelling and grammar errors portray the reviewer more like an average person than a highly skilled marketer with a vested interest. Therefore, the presence of spelling and grammar errors could be interpreted either way, negating their use as a cue in evaluating product reviews. Similarly, research suggests that negative reviews about a product are used by consumers to infer that the review is genuine (Kanouse 1984, Kanouse & Hanson 1972). The current study finds that negative reviews are not used by consumers in evaluating product reviews. This research examines a total of 33 cues and finds that only 22 are actually used by consumers. The 11 cues that are not used by consumers are considered in greater detail above in the Discussion section. Website marketing managers would be wise to focus their attention on the 22 cues that are actually used by consumers in evaluating product reviews and avoid getting sidetracked by the 11 cues that are not used (see Appendix B).

In terms of cues that are used by consumers in evaluating online reviews, the findings regarding authenticity cues may be very useful to marketing managers. While it may be challenging to eliminate fake reviews, encouraging reviews that are perceived as authentic may be more practical and perhaps intimidate fake reviews. As explained by Wu & Qiu (2023) based on their empirical research, ensuring a larger proportion of authentic feedback creates a natural barrier to prevent illegitimate reviews.

The current research identified seven cues related to Review Detail and four cues related to Validation that support the evaluation of a review as credible. In terms of Review Detail, the current research suggests that consumers value specific, in-depth and balanced reviews offered by other consumers with sufficient knowledge in the product category to provide brand comparisons and who are perceived to have experience with the brand as a result of pictures and videos of the product in use. Marketing managers should encourage reviewers with regard to each of these aspects. Furthermore, prospective customers appreciate reviews that are updated over time. Currently, customers who make a purchase are contacted with a request to provide a review of their purchase. Similarly, resellers could proactively follow up with verified online reviewers after a designated period of time (which may depend on the nature of the product in terms of expected frequency of use) to solicit

updates regarding the reviewer's experience.

Manufacturers and resellers can also work together to encourage genuine, validated reviews. Verified buyer badges to reviewers who purchased the brand through their website, requiring reviewers to use their real names, requiring both pros and cons for each review, measuring as well as reporting the extent to which consumers found each review to be helpful, and identifying the percent of helpful reviews for each reviewer (not to be confused with star ratings or aggregated measures) can help in this regard. Reviewer identity in terms of age, gender, and location (Verma et al. 2023) will also instill trust that a review is credible. Furthermore, Verma et al (2023) suggest that online sites promote the credible reviews by verified reviewers on the first page to aid consumers in finding them more quickly. If each of the cues regarding authenticity in terms of both Review Detail and Validation can be developed and encouraged, consumers may more readily disregard the reviews they perceive as fake and rely instead on those they perceive as credible.

Finally, as suggested by Banerjee (2022), web platforms could remind reviewers regarding ethical and social responsibilities before the post can be submitted, emphasizing the importance of presenting an accurate description of the product's performance as well as the user's experience.

7.3. Limitations and future research

Future research could further investigate the insignificant relationship between the negative attitude toward the website and the completion of purchase. While our research inferred a consumer decision-making process that suggested a negative attitude toward the website would likely impede purchasing, we offered the explanations that customers who form a negative attitude toward the website may still purchase the brand (1) from a different website or (2) from the same website for other reasons such as price, shipping costs, delivery, retargeting, etc. As explained by Batra & Keller (2016), the digital era has greatly expanded the consumer decision journey by adding multiple stages to the process along with the potential for moving forward or backward across stages. Additionally, it allows for consumers to expand the information search stage horizontally by first searching online, then visiting stores to see the product first-hand, and finally placing the order online on one of multiple reseller websites. The link between attitudes toward websites and purchasing behavior within the greater framework of the expanded decision journey would be worth exploring. In fact, one of the questions we asked at the start of our survey was an open-ended question asking the respondent to describe the situation when they used online reviews to evaluate the selection of an electronics good. It is interesting to note that over 58% of the respondents explicitly or implicitly mentioned the use of multiple websites for information search. As an example, one of the respondents described how he or she searched product information from multiple resources, searching for "user and tech reviews first" to see "what brands people were recommending and liking," then "narrowed it down to a few brands and searched each of those brands on multiple sites, like Amazon, Best Buy, etc. to read user reviews," and then compared and shopped in different online platforms for the best prices. The amount of search activity and the number of options considered is also interesting to investigate.

The cues for both authenticity and inauthenticity identified in the current research can be further validated by testing the factor structure using a new data sample. The current research was exploratory with regard to the factor structure of the cues. Future research can examine the data using confirmatory factor analysis.

Looking forward, as market researchers continue to explore the cues that consumers use to evaluate an online review as either fake or credible, new cues may emerge and could be added to the investigation. An example of a potential cue that could affect consumer evaluations is the device used by the online reviewer to post a review. While earlier research proposed that due to ease and accessibility, customer reviews

posted using mobile devices (as opposed to non-mobile channels such as desktop computers) would lead to greater review volume, changes in review distribution patterns or average review ratings (Mariani, Birghi & Gretzel 2019), more recent research contradicts those suggestions and yet supports that the device used may still play an important role. More specifically, Kim, Lee & Mariani (2021) found that mobile devices do not influence whether or not a review gets posted but rather when they get posted. That is, mobile devices are more likely to be the channel of choice when customers are eager to post quickly based on either being very satisfied or very dissatisfied (Kim, Lee & Mariani 2021). It is possible that responses posted quickly support that the reviewer has actually experienced the product and are therefore more likely to be perceived as authentic. Online reviews posted using mobile devices may also be less likely to overlap other reviews. If future empirical results bear out this finding, then online review sites would be advised to indicate the type of device used by the reviewer at the time the post was made. A post that indicated a mobile device was used might lend credibility to the review.

One of the measurement items for the Inducement scale was related to compensation. More specifically, if a reviewer revealed that s/he was paid or compensated for the review, respondents identified this as a red flag that the review was fake. However, no differentiation was made between mandatory versus voluntary disclosure on the part of the reviewer. In the future, this item could be modified to two items, one for mandatory disclosure and one for voluntary disclosure. Cui et al (2022) report that mandatory disclosure has a small but positive impact on product sales, while voluntary disclosure has a small but negative impact. It is not known whether voluntary versus mandatory disclosure affects discounting the reviewer's opinion.

Future research could also expand the structural model to consider potential moderating variables. For example, the level of consumer expertise with the brand or product category could attenuate the effect of fake product reviews, because such consumers can evaluate the information and make a decision using their own knowledge and experience and would be less influenced by a single fake review. Recent research by Chatterjee et al. (2023) explains that product knowledge helps consumers avoid deception since they are better prepared to identify incorrect or deceitful information in product reviews. In addition, individual trait propensity for skepticism could be a plausible moderator. A consumer with a high propensity for skepticism may be more inclined to perceive a product review as fake compared with the other consumers. Another moderator which could be investigated is the nature of the product. For example, in their investigation of credible online reviews, Roman, Riquelme and Iacobucci (2023) find differences between search goods and experience goods, suggesting that credence goods might also be investigated.

Future research could also incorporate control variables. For example, the role of culture as it pertains to online reviews may be interesting to explore. Recently, Mariani and Matarazzo (2021) provided empirical evidence that greater cultural distance between a service provider and the customer produces lower levels of customer satisfaction. Therefore, the greater the number of international customers who are culturally different, the greater the number of negative reviews that may be expected. One might wonder then about the impact of perceived cultural distance between the online reviewer and the consumer. For example, Amazon.com has historically distinguished between US reviews and global reviews making it possible for customers to see from which country a review was posted and therefore be able to assess the cultural distance between themselves and the reviewer. Does greater cultural distance lead the consumer to discount the review (recognizing that the online reviewer may simply have different expectations and experiences) or does it lead to greater suspicion of the review? In other words, should cultural distance be considered as a control variable – or as a cue used by consumers in evaluating review authenticity? In the first instance, cultural distance could lead the customer to discount the review regardless of perceived authenticity,

therefore suggesting that cultural distance should be measured as a control variable. In the second case, cultural distance could be used as a cue. Some support for this effect is demonstrated by [Harrison-Walker \(1995\)](#) who found a three-way interaction effect between provider nationality, supplemental information, and customer nationality. It may be that in the absence of significant additional credibility factors (such as in-depth and detailed reviews, updated reviews, and reviews that include pictures and videos), when *perceived* cultural distance is high between the reviewer and the consumer, reviews may be more likely to lead to discounting the reviewer's opinion and arousing suspicion of the review.

The current research identified eleven cues that consumers rely upon to evaluate an online review as fake. The survey question asked respondents to indicate the extent to which each cue led to suspicion of the review as fake. Thus, the cues were implicitly tied to suspicion within the question. Alternatively, the list of cues could be used in future *experimental* research to develop a series of online reviews for a fictional product and then examine respondent attitudes and intentions toward the brand, the website, and purchasing.

In this research we did not differentiate between reviews perceived by consumers as fake which are indeed fake versus those reviews perceived by consumers as fake which are in fact genuine. Further study could explore the differences between actual fake reviews and real reviews that are perceived as fake, together with the implications of these two types of reviews.

Finally, we were able to identify two factors comprised of 11 items that consumers use to signal that an online review is credible. The credible cues would be necessary to experimental research for presenting a mix of credible and fake reviews as would be the case in a real product evaluation situation. The question arises whether perceived credibility of online reviews significantly affects brand purchases or whether perceived credibility is instead 'expected' by consumers and only in the presence of perceived inauthenticity are sales negatively affected.

CRediT authorship contribution statement

L. Jean Harrison-Walker: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ying Jiang:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A.: Measurement items for suspicion and outcomes

Suspicion	(Adapted from Zhang et al. 2016)
Truth	
TRU1	I could not depend on getting the truth from these online reviews.
TRU2	The online reviews were not generally truthful.
TRU3	The online reviews did not reflect a true picture of the brand.
Motivation	
MOT1	The online reviewers cared mostly about getting me to buy the brand.
MOT2	Most of the online reviews were intended to mislead.
MOT3	The people writing the online reviews were up to something.
Identity	
IDY1	The reviewers writing the reviews are not necessarily real customers.
IDY2	The reviewers wrote the reviews pretending to be someone else.
IDY3	Different reviews were posted by the same person under different names.
	Scale Alpha = 0.916 Study 1/0.938 Study 2
Credibility	(Appelman and Sundar 2016)
CRE1	Accurate
CRE2	Authentic
CRE3	Believable
CRE4	Reputable
	Scale Alpha = 0.932 Study 2
Discounting Reviewer Opinion	
DRO1	After reading the online product reviews, I place less value on the reviewers' opinions.
DRO2	The online product reviews lead me to discount the opinions of those reviewing the product.
DRO3	The online product reviews left me with an unpleasant feeling toward the reviewers.
DRO4	I feel that cannot place much value on the reviewers after reading the online product reviews.
DRO5	These kinds of product reviews cannot be counted on when I'm evaluating a brand.
	Scale Alpha = 0.902 Study 1/0.937 Study 2
Negative Brand Attitude	
ATB1	Based on my perception of the product reviews, I think less of the brand.
ATB2	After reading the online reviews, I no longer found the brand to be appealing.
ATB3	The online reviews made me dislike the brand.
ATB4	I no longer favored the brand after reading the online product reviews.
ATB5	I do not like brands when these kinds of product reviews are posted.*
	Scale Alpha = 0.923 Study 1/0.939 Study 2
Negative Attitude Toward the Website	
AWS1	After reading the online product reviews, I came to distrust the web site hosting the product reviews.
AWS2	I cannot trust the website because it allows these product reviews to be posted.
AWS3	Based on my perceptions of the online product reviews, I do not believe I can count on the information posted on this website.
AWS4	My perception of the website is greatly reduced given the online product reviews posted there.
AWS5	I do not like websites that allow these kinds of product reviews to be posted.
	Scale Alpha = 0.895 Study 1/0.937 Study 2
Not Completing a Purchase	
PUR1	Based on my perception of the online product reviews that I read, I was less likely to purchase the brand.

(continued on next page)

(continued)

Suspicion	(Adapted from Zhang et al. 2016)
PUR2	Reading the online reviews caused my interest in purchasing the brand to decline.
PUR3	I was less likely to buy the brand having first read the product reviews.*
PUR4	My perception of the online product reviews reduced my interest in purchasing the brand.
PUR5	The online product reviews made me far less likely to purchase the brand.
Scale Alpha = 0.923 Study 1/0.957 Study 2	

* Items removed during scale purification process.

Appendix B:. Measurement items for evaluating reviews

Cues Indicating Fake Online Reviews	
Perceived Reviewer Inducement	
IND1	Reviews that disclose that the reviewer was given a product discount for providing the review.**
IND2	Reviews that disclose that the reviewer has been paid or compensated for the review.
IND3	Reviews that indicate the review was 'submitted as part of a sweepstakes entry'.
IND4	Reviewers that ask the buyer to buy, click, or forward the review.
IND5	Reviews submitted by experts who specialize in the product category and have a vested interest in your brand choice.*
Review Imbalance	
IMB1	Reviews that are extremely positive.*
IMB2	Reviews that are extremely negative.*
IMB3	Product reviews where the positive reviews greatly outnumber the negative reviews.*
Review Overlap	
OVR1	Reviews in the review set that are time stamped too close to each other, even though posted by different reviewers
OVR2	Multiple reviews in a set of reviews that sound or appear similar to each other.**
OVR3	Product reviews with repetitive wording with other reviews in the review set.
Review Composition Nonconformity	
CMP1	Reviews that contain a significant number of spelling errors.*
CMP2	Reviews that contain improper grammar.*
Reviewer Implausibility	
IMP1	Reviews posted by reviewers with nonsensical usernames like x827bdjn19
IMP2	Multiple reviews of the same product using nonsensical usernames (such as x827bdjn19).
IMP3	Reviewers that use excessive cliches (such as, "This is the last x you will ever have to buy!")
IMP4	Reviews where phrases have been copied directly from the product description page
Cues Indicating Credible Online Reviews	
Reviewer Validation	
RVL1	Reviews where the reviewer is identified by his/her real name.
RVL2	Reviews that include the photo of the reviewer.*
RVL3	Reviews where the reviewer's rating by consumers is disclosed (for example, where top reviewers might get five stars).*
RVL4	Reviews that indicate the reviewer has a high 'Helpful Review' count (e.g., positive responses to 'Was this review helpful?')
Review Detail	
DET1	Product reviews that are very in-depth.
DET2	Reviews that are very specific in detail.
DET3	Reviews that are very long.*
DET4	Reviews that appear to be written by other customers who know a lot about the product category.
DET5	Balanced or two-sided reviews (showing both pros and cons of the product in a single review).
DET6	Reviews that have been updated since their original post to provide additional information or feedback.
DET7	Reviews where the reviewer provides comparison to similar models and features.
Purchase Validation	
PVL1	Reviews that show a video of the product in use.
PVL2	Reviews that show a picture of the product in use.
PVL3	Reviews that provide 'Verified Buyer' badges for reviewers to support that the reviewer has actually purchased the brand.
Negative Reviews	
NEG1	Reviews that are unfavorable to the product.*
NEG2	Negative reviews about the product.*

* Items not supported in Study 1.

** Items not supported in Study 1 but supported in Study 2.

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